

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 102 — Past Questions & Materials
by Renance

Course: PEE 102
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 100

This study pack was compiled and restructured by Renance for PEE 102 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 3 — FUPRE-PEE 102 Past Questions - Basic Oil and Gas Industry Operations and Processes

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING

2022/2023 First Semester Examination

PEE 211: Introduction to Oil and Gas Industry

Time Allowed: 1hr. Date: 27/05/2023

QUESTIONS

Answer ALL Questions.

1ai), State the main five sectors of the petroleum industry as classified by the American Petroleum Institute (API). |

ii). Why is the petroleum industry considered to be so complicated?

b). List the seven main aspects of the oil and natural gas value chain?

2ai). How can we determine where to drill and predict what we will find BEFORE we start drilling?

li). What would one do to maintain a healthy petroleum company?

b). What is the difference between a "Refinery" and "Oil Refining"?

3ai). Companies in the Petroleum industry can be subdivided into four major categories. State them.

ii). State in no particular order, the top 5 largest world oil companies by reserves and production.

b). Differentiate between "Refined Product" and "Petrochemical Product".

4ai). What do you understand by the term "The Seven Sisters"?

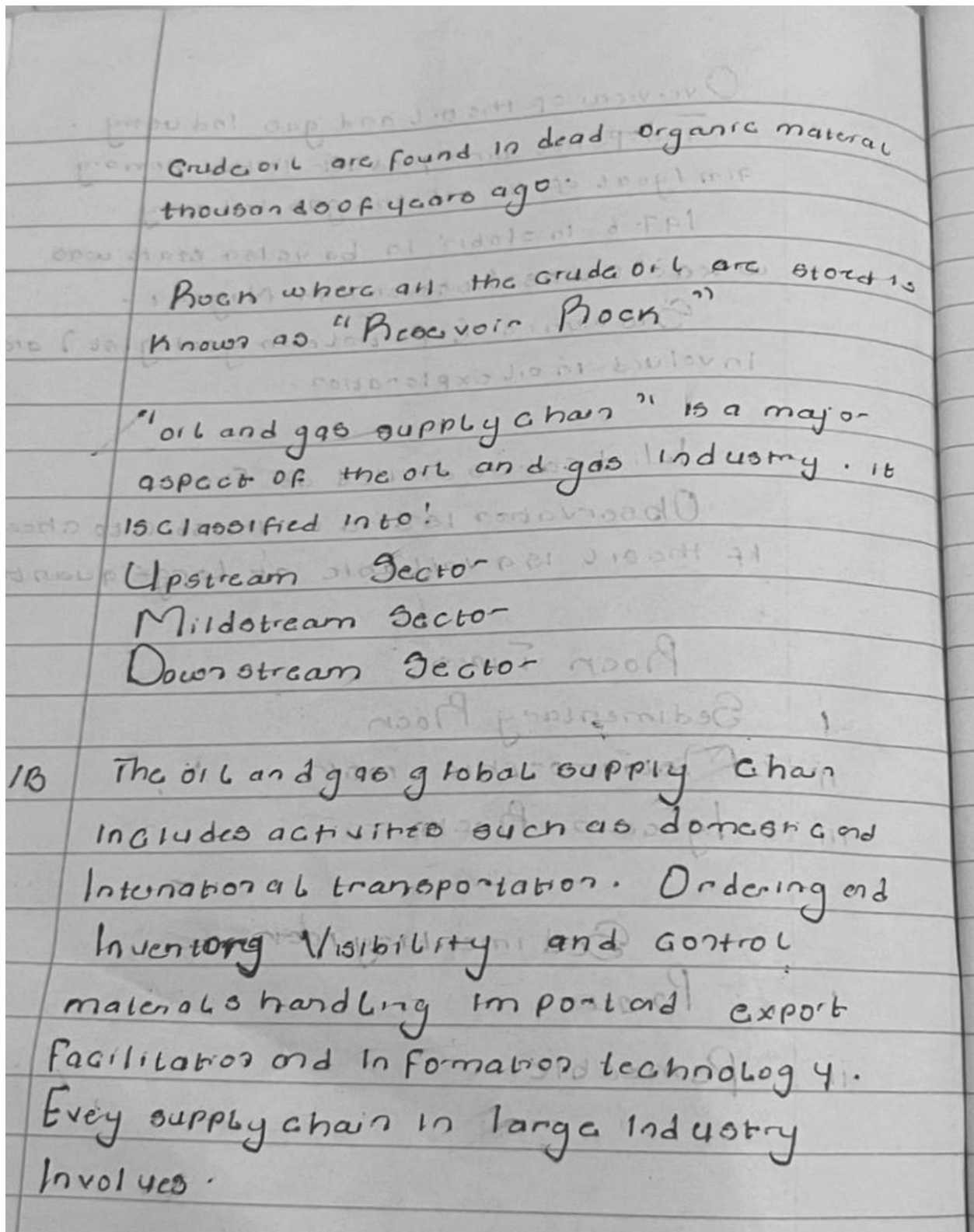
li). What do you understand by the term "New Seven Sisters"?

b). What are Service Petroleum Companies?

Paper 2 of 3 — PEE 102 Note By THEOCO Second Semester

Level 100 • Second Semester • Handout

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- i Configuration
 - ii Management
- Continuous Improvement of sequential set of Operation that includes multiple Parties.

The goal in supply chain management is to deliver maximum service to the customer at the lowest cost possible.

Upstream	Midstream	Downstream
oil and gas exploration Production	Transport and Storage	Product Preparation and Usage
Off-shore oil and gas platform	Process and storage	Refining.
Onshore Pump Jack	Transport	Distribution, sales, Marketing/ Retail.

Onshore are Land Platforms for oil exploration and mining

Upstream Sector Company

- i Total
- ii Mobil
- iii Shell
- iv Chevron
- v Agip

Midstream Sector Company

- i Pipeline transportation
- ii Tankers

Midstream Companies in Nigeria

Matrix

Oando

Com oil

Product preparation and usage are done by the Downstream sector

Distribution, sales, marketing and retail
Petroleum Production is done by the downstream sector.

Upstream Sector

Also known as the A and B it is
consisted of Process and operation that
involved searching for potential underground
or underwater crude oil and natural gas
field.

Field is a wide location where crude oil
is located. Pockets of oil wells are
located in a field. Drilling of exploratory
wells and subsequently drilling and operating
the well that recovers and bring the crude oil
or raw natural gas to the surface. In
recent years there is an evident shift towards
the inclusion of unconventional gas as
part of the Upstream Sector. This also
affects the development in processing and
transporting liquefied natural gas (LNG).

Midstream Sector

This is usually combined in the literature with the downstream sector. This segment in the supply chain involves in the transportation, storage and marketing of oil and gas product. Transportation options can vary from small connector pipelines to massive cargo ship making trans-ocean crossing depending on the commodity and distance covered. When discussing the transportation of oil and natural gas, most oil can be transported in the current state while the natural gas must be liquified or compressed.

DOWNSTREAM SECTOR

It encompasses the refining, processing, distillation, purification and purification before turning into usable, sellable and consumable product. For example fuel, raw chemicals and finished products etc. All the mentioned services transform crude oil into usable product such as gasoline,

Fuel oil and Petroleum based Product.

Retail marketing activities help move the finished Product from energy Company to retailer or end user.

UPSTREAM SECTOR CHARACTERISTICS

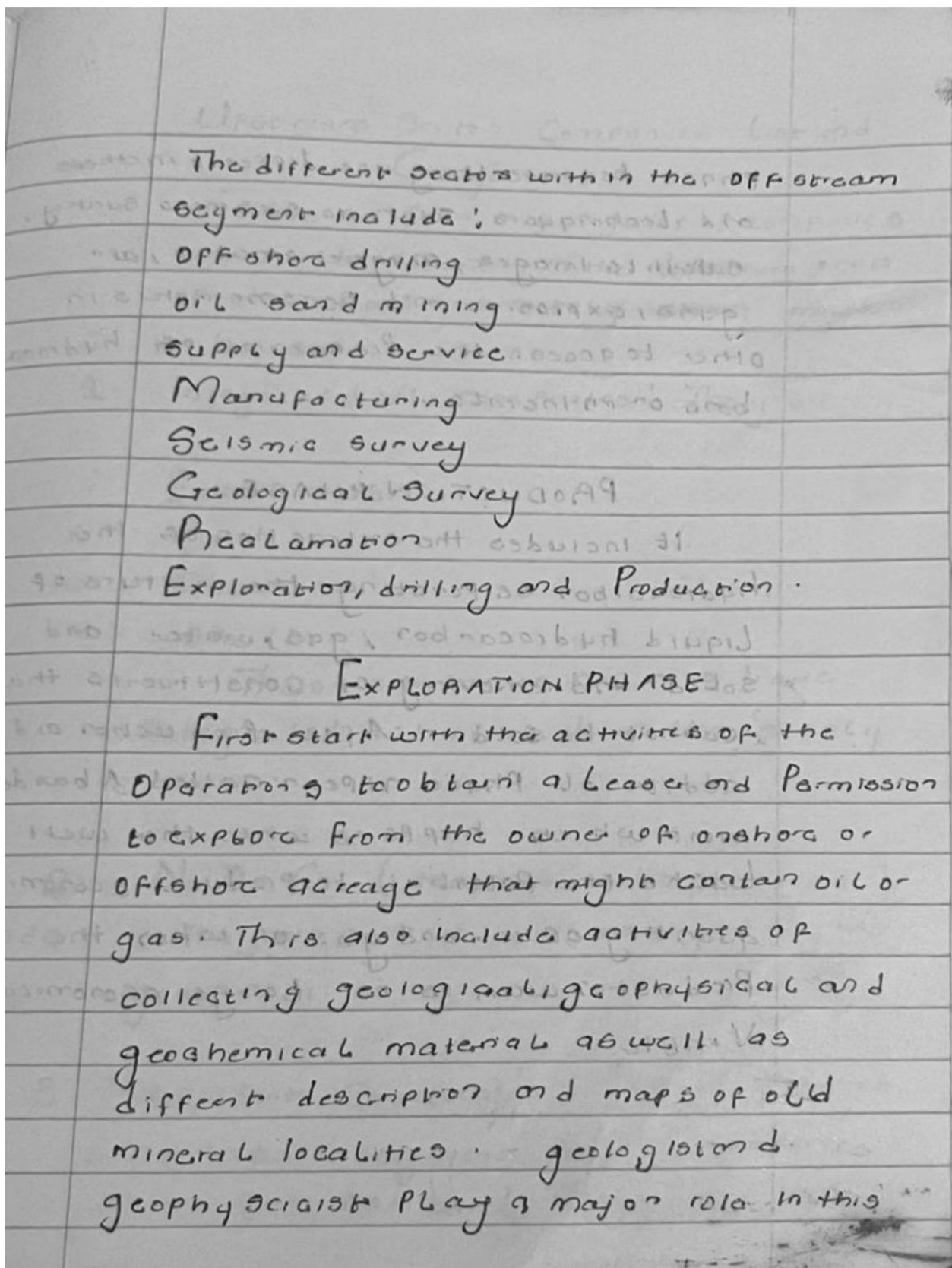
This segment of oil and gas supply chain focuses on Operators around the well. It cares about where to locate them how deep and far to drill the well. How to design, construct and manage them. The four major business characteristics that describes the Upstream sector are as follows;

- 1 They carry high risk
- 2 Give high return
- 3 Highly regulated
- 4 It is affected by global politics
- 5 It is capital intensive.

30/07/2024

OVERVIEW OF THE OIL AND GAS INDUSTRY

Exploration	Drilling / Development	Production
Geological	Field Planning	Well
Prospectivity		Completion
Seismic	Engineering and	Production
Imaging	Construction	Operations
Exploratory	Development	Well
Drilling	Drilling	Intervention
		work-over



Phase because they use different methods and techniques such as seismic survey, satellite images, magnetometers, air guns, explosives and seismometers in order to access the presence of hydrocarbons or minerals.

PRODUCTION PHASE

It includes the extraction of the hydrocarbon separating the mixture of liquid hydrocarbon, gas, water and solid and removing of constituents that cannot be sold. After production an additional phase appear called Abandonment which happens when the well lacks the potential to produce economic quantity of oil and gas or when the production well is no longer economically viable.

Upstream Sector - Companies List and Categorization

There are four major group of companies that can be identified in the upstream sector of the oil and gas supply chain, majors or known as:

1 Integrated oil companies: They are

- i Exxon mobil
- ii BP
- iii Chevron
- iv Shell

These companies can also have integrated activities down the supply chain.

2 N.O.C (National oil companies)

They are owned and managed by the government.

3 Independent Companies: These companies exist in each segment the independence comes from not being integrated into

Overview of the oil and gas Industry .

The Petroleum product is the primary aim of the oil and gas industry .

1976 in Olofin in bayelsa state was where oil was discovered in Nigeria .

Geometric (Petroleum geologist) are involved in oil exploration .

Well got drilled at first

Observation is done on the well to check if the oil is available at large quantity

Rock Types

i Sedimentary Rock

ii Metamorphic rock

iii Igneous Rock

Sedimentary rock

i Porous

ii Permeable

other segment:

4. Oil Field Service Companies: These companies offer services, equipment and different technical skills related with exploring, drilling, testing and producing oil and gas.

XHP (Extra high Pressure)

If your foam began to foam, it is called "Anti Foam".

"Lat. Appearance Temperature" is the point the foam react.

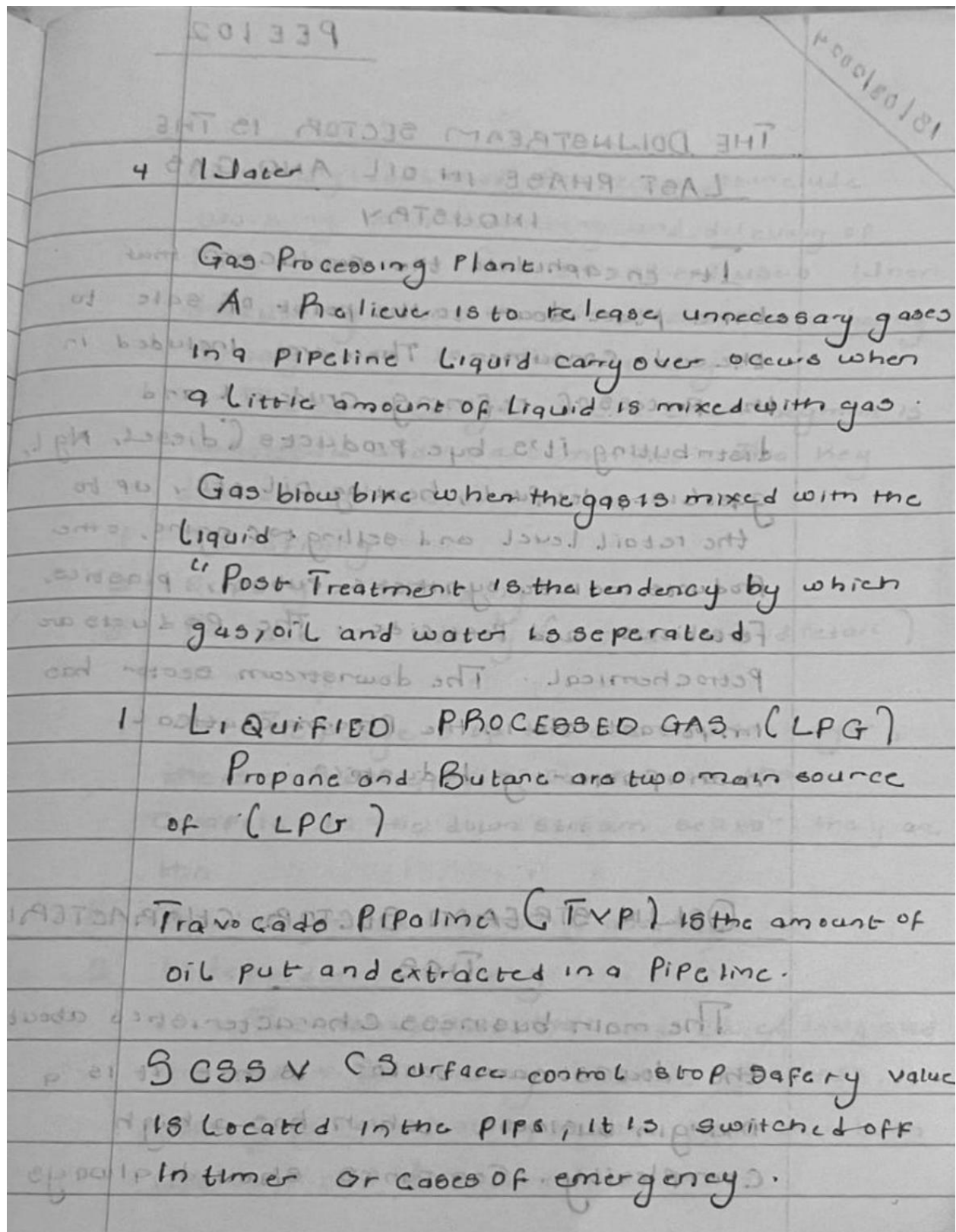
Desalt is the removal of salt

Pipeline process is called the "flow line"

Desand is the removal of sand.

Separation of Oil

- 1 Gas
- 2 Oil
- 3 Emulsion



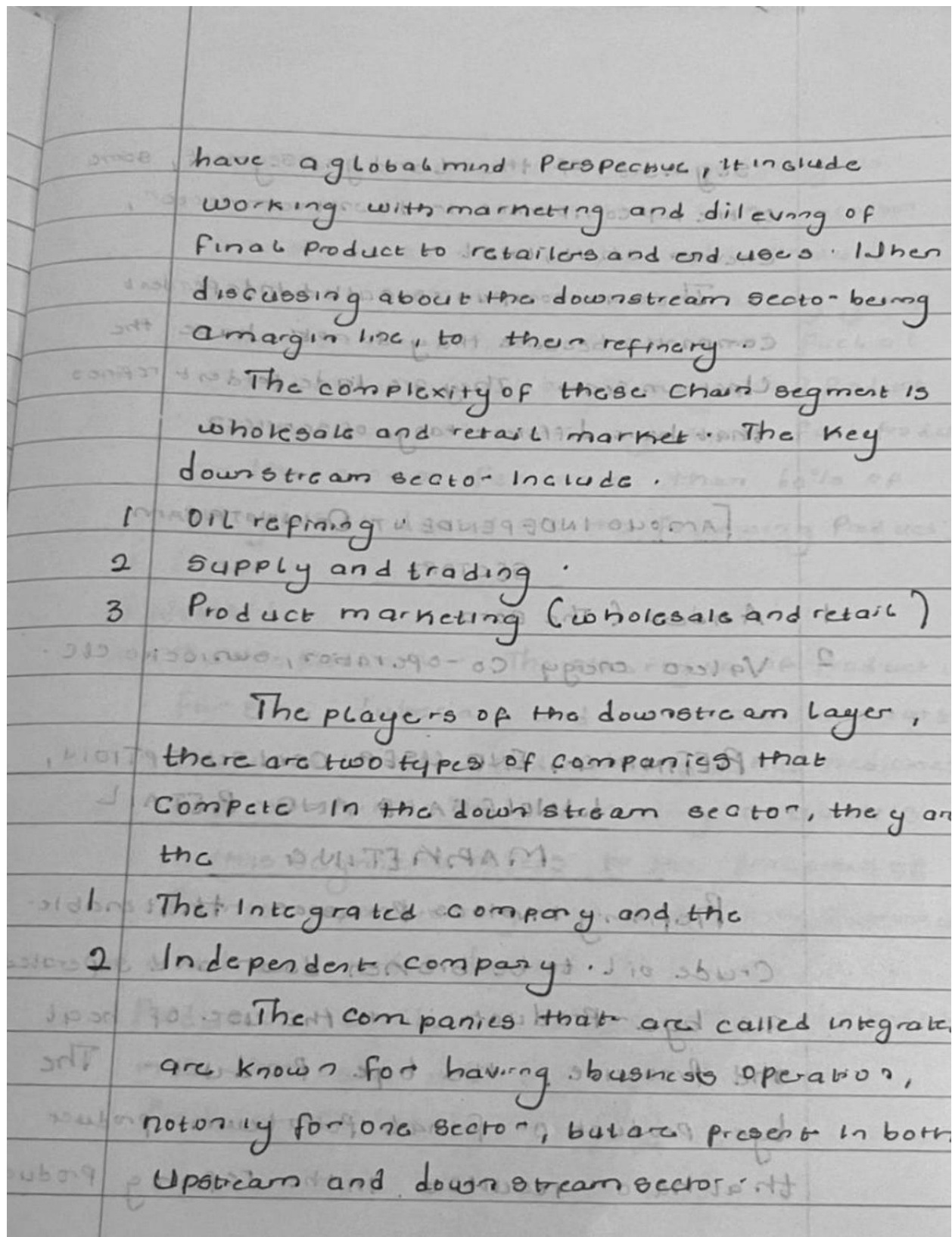
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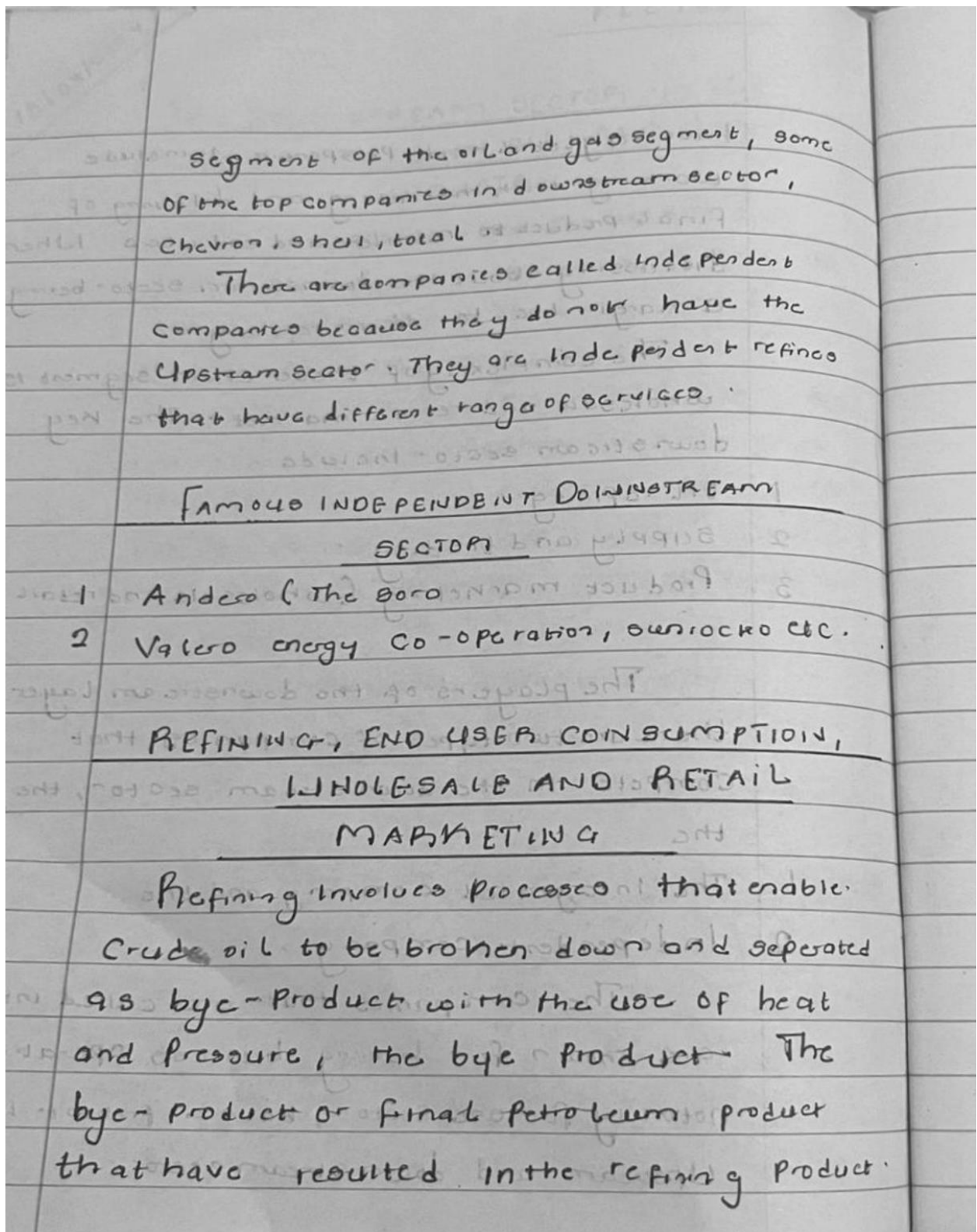
THE DOWNSTREAM SECTOR IS THE LAST PHASE IN OIL AND GAS INDUSTRY

It encapsulates the production that takes place down to the point of sale to the end consumer. There are included in the process of refining crude oil and distributing its by-products (diesel, NGL, gasoline, jet fuel, heating oil etc), up to the retail level and selling to engine, other products include synthetic rubber, plastics, fertilizers and pesticides. The products are Petrochemical. The downstream sector has important role in the pharmaceutical, manufacturing, industries.

DOWNSTREAM SECTOR CHARACTERISTICS

The main business characteristics about the downstream sector is that it is a margin business, which has a high complexity. Companies should always





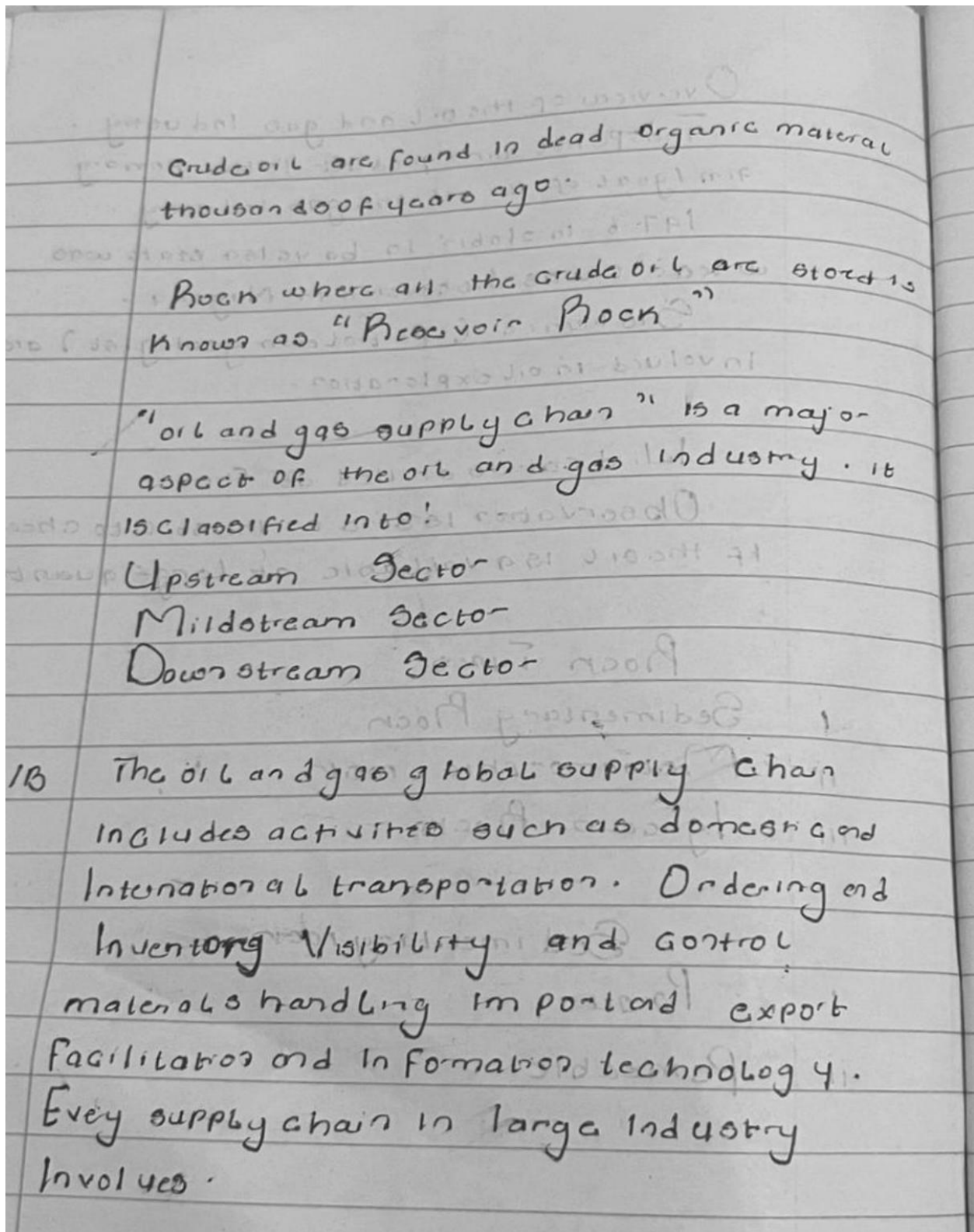
Can be categorised into three major categories -
Liquid Petroleum or NAPDC, hydrocarbon
also known as middle distillate products
Jet fuel and kerosene while heavy group
of carbon atom forms the residual fuel oil.
Gasoline is the most globally used product
of crude oil. Other widely known fuel product
that consist far more than 60% of
the global demand are the following products:

- i Diesel
- ii Jet oil
- iii Marine fuel oil. The other type of product,
for ex; lubricant and waxes are important
and used in other industries such as medicines
and cosmetics. The marketing activities
in this segment refers to the processes of
Promoting, searching and supplying customers
who have internal demand for refined fuel
product who have large wholesale network
that can distribute the different
product to variety of retailers.

Paper 3 of 3 — PEE 102

Level 100 • Second Semester • Handout

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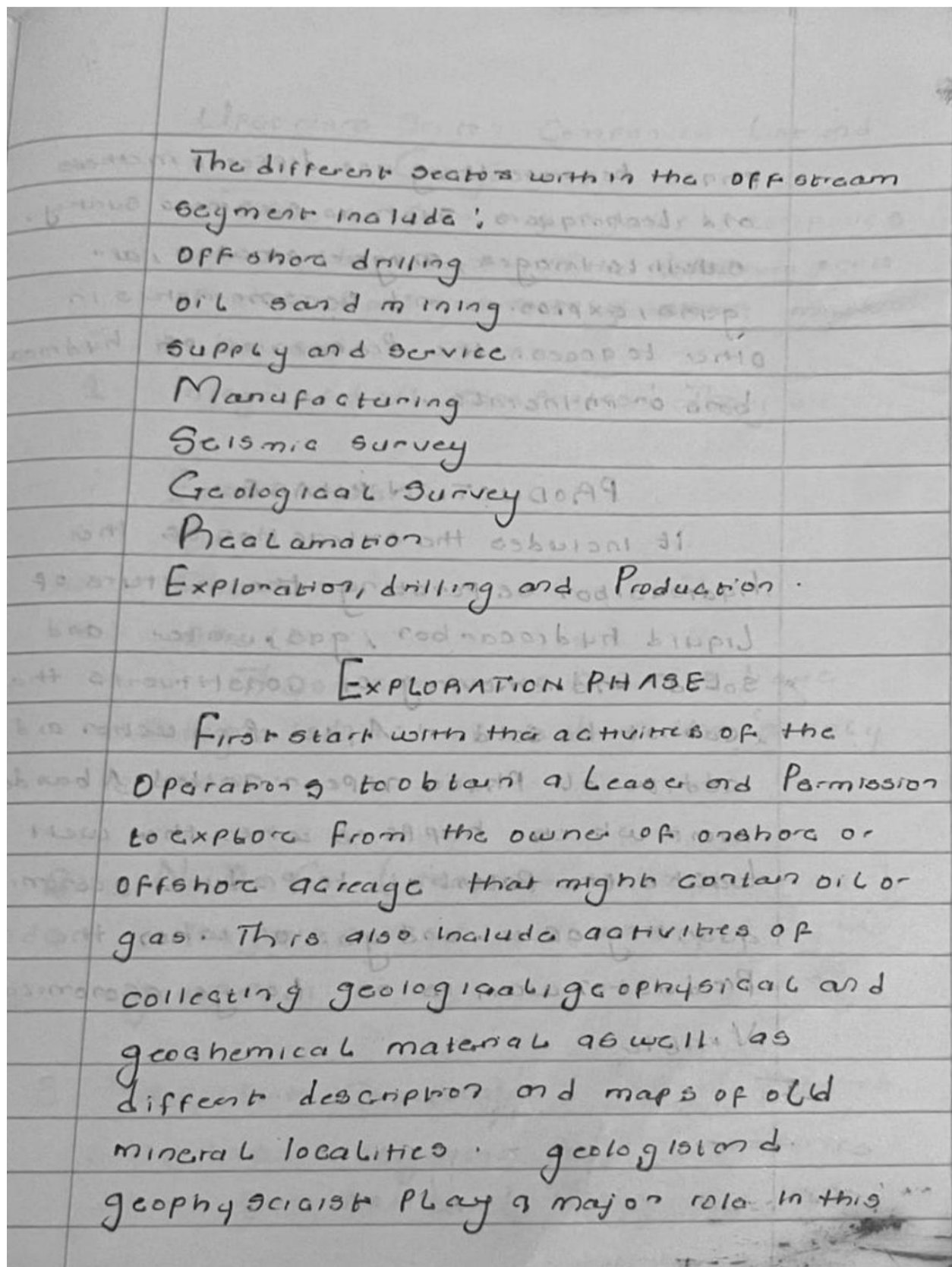
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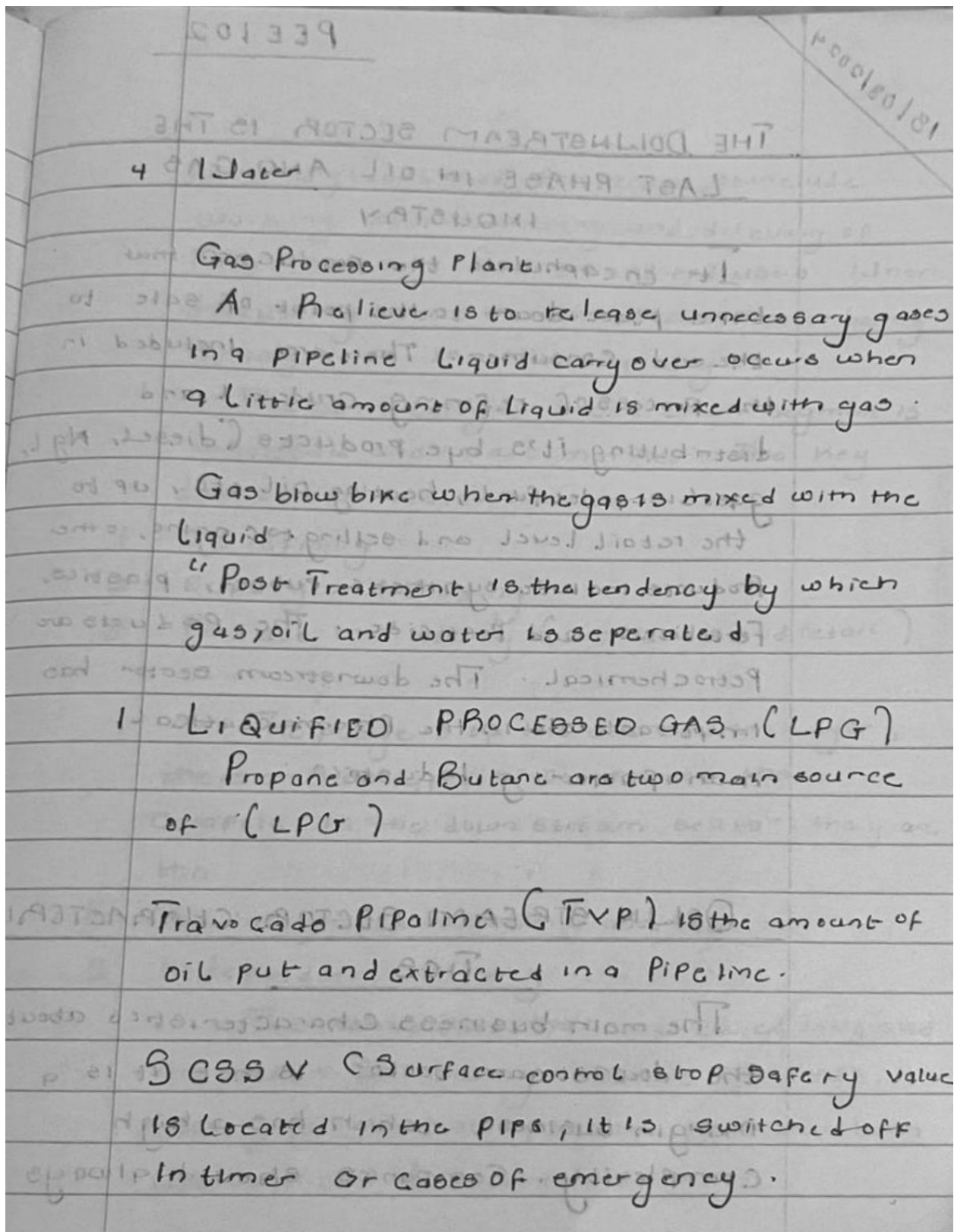
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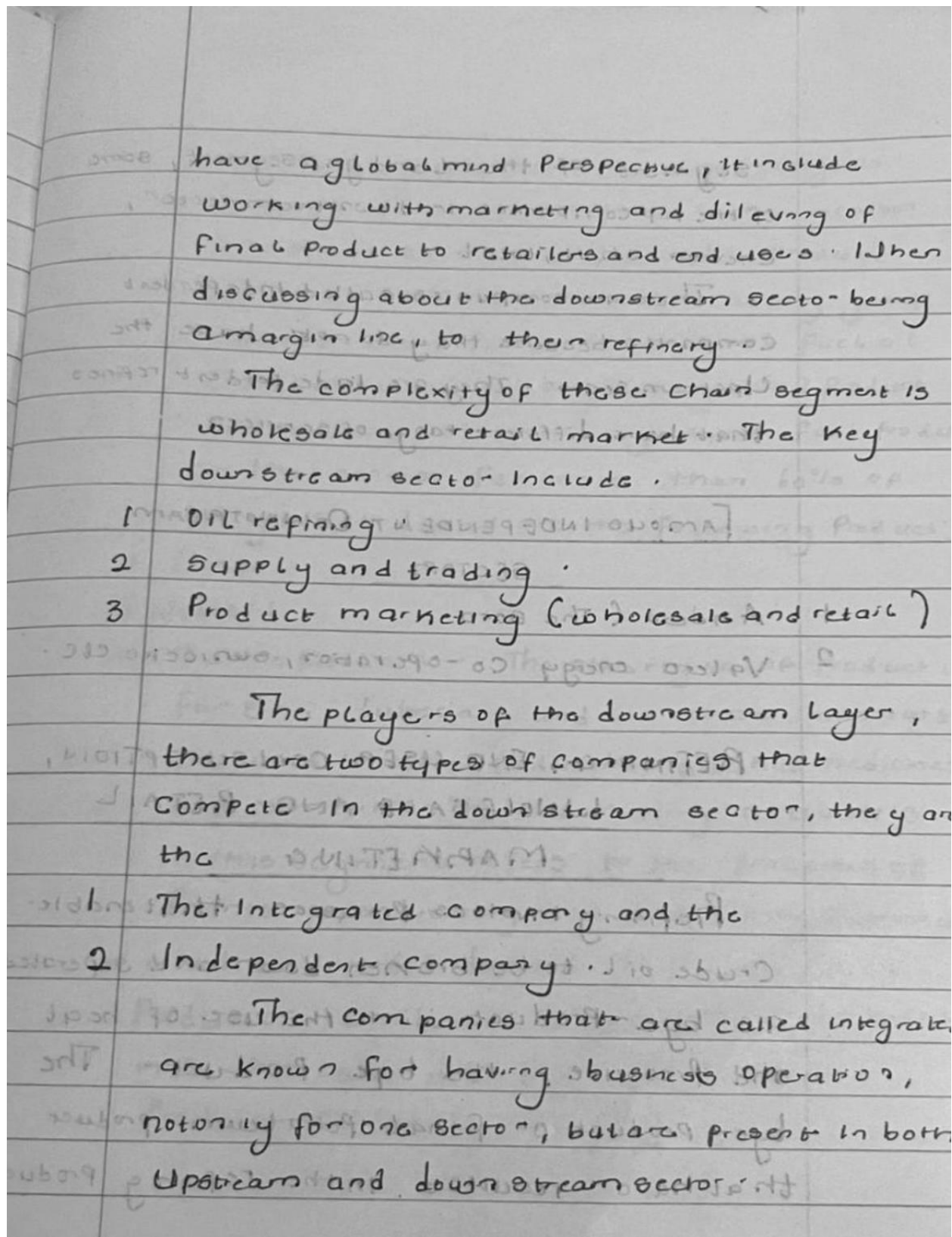
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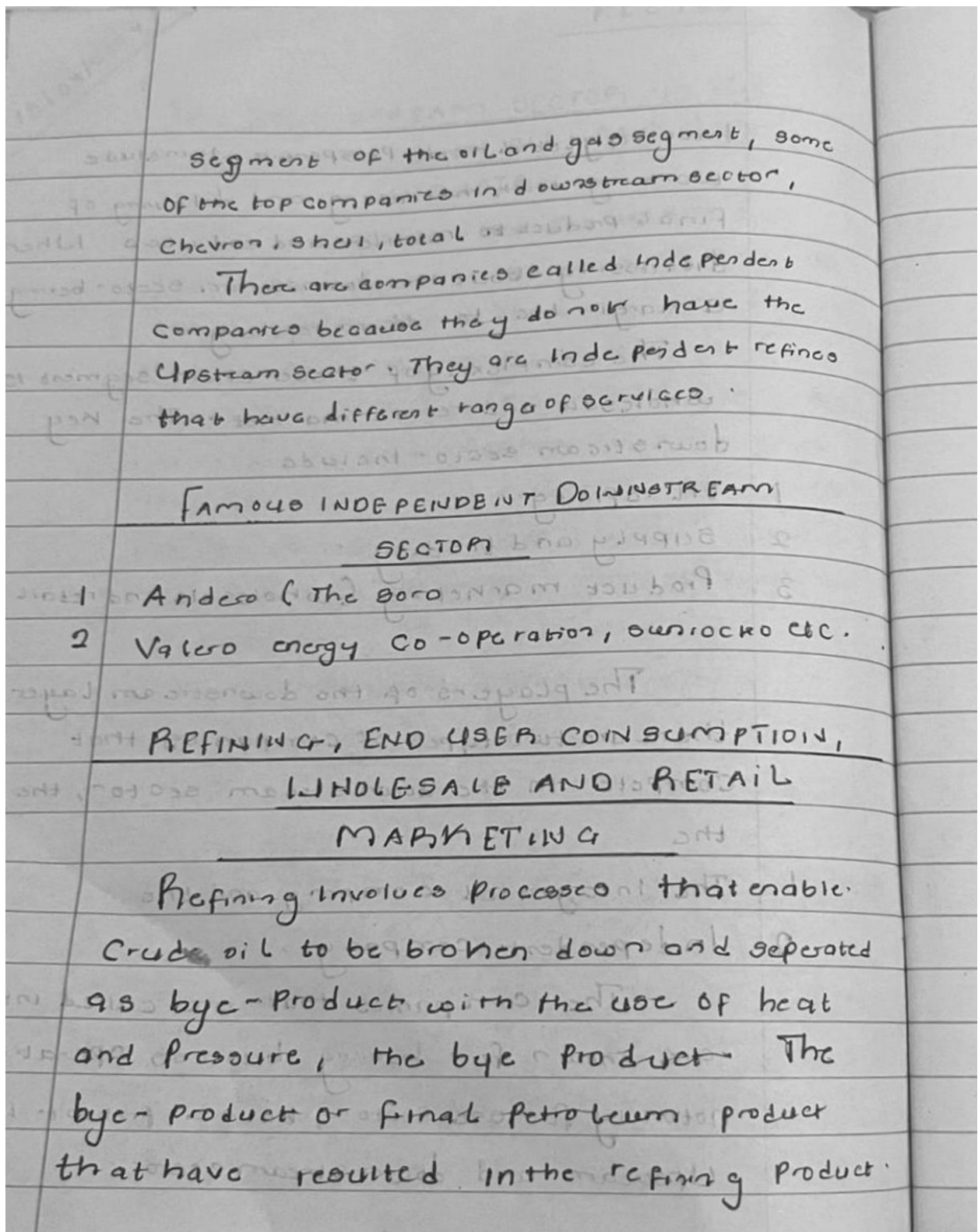
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